

Deep learning for the classification of cassava leaf diseases in unbalanced field data set

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Abstract. Cassava is one of the main sources of carbohydrates in the world. However, the diagnosis of diseases in cassava crops is laborious, time-consuming and requires specialised personnel. In addition, very little research is available on images of cassava leaves taken with mobile phones and under field conditions. Therefore, the study designs deep learning models for the detection of diseases in cassava leaves from photos taken with mobile phones in the field. This study used a dataset of 21'397 images of cassava bacterial blight, cassava brown streak disease, cassava green mottle and cassava mosaic disease from a Kaggle competition. Twelve CNN models have been evaluated by applying transfer learning and data augmentation. Each of the models was trained with uniform samples and class-weighted samples. The results showed that the use of weighted samples reduced F1 score and accuracy in all cases. Furthermore, the DenseNet169 model was outstanding with an accuracy and F1 score of 74.77 % and 0.59 respectively. Finally, the causes that hinder correct classification have been identified. The results reveal that it is still necessary to work on creating a balanced and refined database.

Keywords: Deep Learning · Plant disease · Convolutional neural networks · Leaf disease · Classification

1 Introduction

Manihot esculenta, commonly called cassava, is an ancient crop, which is native to South America [1–3]. However, it is found all over the world, with the greatest popularity in the Americas and Africa. Cassava is a food rich in carbohydrate content, which makes it one of the most important sources of energy in the world [4, 5]. Nevertheless, cassava is reputed to be a crop often produced in the poorest and most remote areas [6]. Additionally, yields in central and west Africa compared to other cassava producing regions such as Asia or Latin America are generally lower [7].

Since 2011, Food and Agriculture Organization (FAO) and other studies have warned about the epidemic effects of the cassava brown streak that attacks in Africa [8]. In addition, studies have also indicated that climate change is expected to increase the incidence of diseases [9].

Examination or inspection by the naked eye for plant diseases usually based on colour changes or the existence of spots or rotten areas on the leaves is the

most commonly used mechanism to detect and classify diseases [10]. However, farmers and inexperienced people have difficulties in identifying diseases based on the symptoms shown as the same diseases may show different symptoms or different diseases may show similar symptoms [11]. Even determining or evaluating a disease by symptoms is difficult if done manually by one or more specialists [12]. In addition, inspection by experts is often costly, time-consuming, slow, subjective and in some cases impractical [13, 14, 16].

Crops can be affected by two types of diseases; biotic and abiotic diseases [17]. Biotic diseases are caused by living organisms, such as bacteria, viruses and fungi. These diseases are contagious and, depending on the case, can be dangerous. On the other hand, abiotic diseases are caused by ecological circumstances, such as climatic conditions, use of chemicals, hail, etc. These diseases are neither infectious nor transmissible and are usually preventable [18].

Disease symptoms in plants can show up in the whole plant structure, e.g. in leaves, stems or roots. However, leaf analysis is more commonly used for disease detection [18].

[14] reviewed the state of the art in plant leaf disease classification, in which it is mentioned that most of the Machine Learning (ML) studies implemented Support Vector Machines (SVM), K-Nearest Neighbours (KNN) and Random Forest. While the main disadvantages of these methods include: low performance, necessity of manual feature extraction [15] and use of segmentation algorithms.

In a real environment, classification of plant leaf diseases is a very complex activity. Different diseases may present the same symptoms and different degrees of disease severity have to be analysed. In addition, we have to deal with a variety of scenarios, different natural lighting conditions, noises in the image, changes in the size of the lesions, etc [19]. Thus, studies based on deep learning algorithms for the detection of diseases in leaves are of interest to the academic field and with prospects for market applications [20, 19, 16].

Several surveys have shown that Deep Learning (DL), essentially Convolutional Neural Networks (CNN), have had outstanding performances with respect to traditional Machine Learning algorithms in the task of leaf disease detection and classification [21, 14, 22, 16].

Several state-of-the-art DL models have already been implemented for leaf disease detection. For example in [23], ResNet obtained the best results in the identification of apple leaf diseases compared to AlexNet and GoogleNet. In [24], 8 types of leaf diseases were identified. Three feature classifiers, SVM, Extreme Learning Machine (ELM) and KNN, were used. Six DL models such as GoogLeNet, ResNet-50, ResNet-101, Inception-v3, InceptionResNetv2 and SqueezeNet were used as feature extractor. This study concluded that Resnet-50 and SVM obtained the best results in performance metrics such as sensitivity, specificity and F1 score. In [16] a comparative study was conducted with 14 crop species and 26 diseases. They concluded that the ResNet-101, Dense-Net201 and Inceptionv3 CNN architectures were the most suitable models for using in standard computing environments, while ShuffleNet and SqueezeNet were the most suitable architectures for mobile and embedded applications.

VGG has also shown outstanding results versus other architectures. Five CNN architectures for plant disease detection and diagnosis, AlexNet, AlexNet-OWTbn, GoogLeNet, Overfeat and VGG, were used in [25], where VGG outperformed all other models. Additionally, AlexNet and VGG are the most widely used models in the recent literature on leaf disease detection. Then, they are followed by ResNet [22].

In [21] a review on detection and classification of plant diseases by DL was conducted. About 40 researches were analysed, and it was found that most of the researches used the PlantVillage dataset to evaluate the accuracy and performance of DL models. This, PlanVillage is still used in various researches because, as mentioned in [14, 26–28], it is currently difficult to access a dataset with adequate size and diversity. In other cases, as in [29], the datasets used contained plant images that were difficult to obtain in real life. Finally, a relevant fact is that, according to [18], no studies have been found on disease identification with acceptable accuracy in the real-world scenario.

Some studies have analysed models trained with data generated in controlled environments[30, 16]. The results showed a significant deterioration in detection accuracy when the model is tested with data taken in real conditions. For example, [31] tested AlexNet and GoogLeNet models with a public database of 54306 images collected under controlled conditions to identify 14 crop species and 26 diseases. They obtained a maximum accuracy of 99.35%. However, the accuracy of the model was considerably reduced when tested under different conditions because the training was performed under ideal conditions, such as single leaves, face up, on a homogeneous background.

The scientific community has revealed the need to create models that are robust to different lighting conditions, i.e. the datasets must contain images at different times of the day. Furthermore, the models must be efficient for different real working environments [32, 21]. Other authors also mention that to avoid biases, data acquisition should occur in several farms to gain robustness to different field maintenance conditions and should be captured with different devices but with similar specifications to those used during the implementation phase [33].

Variability in a data set must also be ensured by capturing various plant disease symptoms. Symptoms are known to be the result of the interaction of planting, disease, and environment [34]. Symptoms due to a plant disease depend on the stage of disease development [32], the presence of other diseases at the same time and presence of similar symptoms due to other diseases. Whereas symptoms due to the environment depend on other factors such as temperature, humidity, wind, soil conditions and sunlight [14].

The use of mobile phones has become widespread even in marginalised sectors, thanks to their easy accessibility, low cost and usefulness. Moreover, some research has already been conducted in which the mobile phone camera is used as an image capturing device [35]. This makes it feasible to use plant disease classification models through apps as proposed in [20].

However, the quality of a classification model to work in real environments is not assured with the use of mobile cameras. The model may present weaknesses if it is trained only with images taken with the optical axis perpendicular to the leaf plane or by extracting leaves for picture taking [18]. Both factors would bring us closer to experiments under controlled conditions.

In [18] analysed about a hundred researches on identification and classification of plant diseases through leaf images, in which it was reported that less than 10% of the papers have used large databases ranging in thousands. It is therefore evident that future research efforts should aim at collecting large and diverse datasets. Similarly in [22], a review of the literature concerning plant disease recognition in images using CNN was carried out, in which 130 manuscripts were reviewed, however only 2 of them had cassava as the crop analysed.

In [36] solved a multi-class classifier problem and tested the Inception V3 model with a maximum overall accuracy of 93%. Two datasets were formed: an "original cassava dataset" with only 2756 images of whole cassava leaves, and a "leaflet cassava dataset" where 15000 leaflets were manually cropped from the first dataset. The accuracy obtained were slightly higher for only three of the five diseases studied with the second dataset. Data were collected with a Sony Cybershot 20.2-megapixel digital camera under uncontrolled conditions. While in [37] a detection problem was solved and the MobileNet & SSD was tested. In addition to the 6 classes of images implemented in [36], an additional nutrient deficiency class of 336 images were included in this work. A 32% drop in F-1 score was reported for pronounced symptoms in real-world images.

It is noted that disease detection using plant leaf images is a popular field of research. Moreover, there is a clear tendency to use DL techniques because of their better performance compared to ML techniques. However, few investigations have cassava as a subject of study. In addition, most of them have a limited amount of images or were performed under controlled conditions.

The importance of this study lies in addressing the problem of disease detection in cassava leaves using the most extensive database of images taken by mobile phones in the fields and under field conditions. The paper evaluates twelve of the most popular DL models in disease detection. Additionally, the effect of using class weights during training for unbalanced data is studied. Finally, we highlight the analysis of prediction error which is not detailed in any of the papers analysed.

2 Methods

2.1 Dataset

This study used a dataset of 21'397 images labelled by experts from the National Crops Resources Research Institute (NaCRRI) in collaboration with the AI lab at Makerere University [38]. In addition, the images were collected by farmers in their own crop fields in Uganda using smartphones [39]. Therefore, the collected images faithfully represent the actual conditions of the application.

The dataset consisted of four disease categories or a fifth category corresponding to healthy leaves. The diseases under study were: cassava bacterial blight (CBB), cassava brown streak disease (CBSD), cassava green mottle (CGM) and cassava mosaic disease (CMD). However, the data was unbalanced as shown in Figure 1.

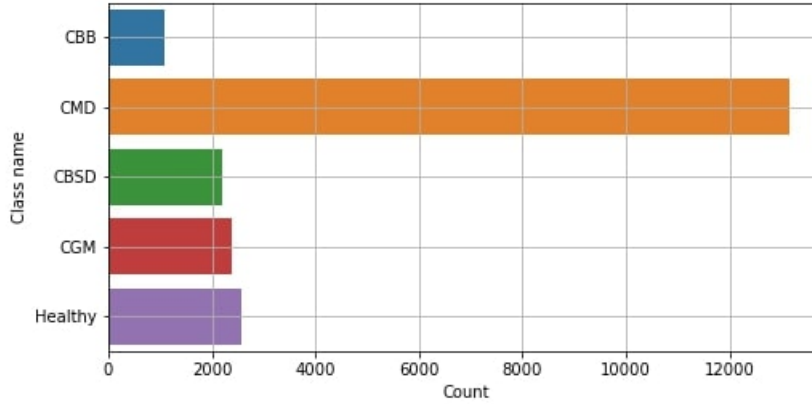


Fig. 1. Data distribution.

CMD was the class with the highest representation (61.5%), while significantly lower were the classes: Healthy (12.0%), CGM (11.2%), CBSD (10.2%) and finally CBB (5.1%). Additionally, Figure 2 shows some samples of each class.

2.2 Pre-processing

The images were resized from 800 by 600 to 224 by 224 pixels in order to reduce the number of model parameters, thus avoiding falling into over-fitting. In addition, data augmentation techniques [40] were used to overcome the class imbalance. Therefore, random rotations from 0° to 45° , random brightness changes between 75-100%, random zooming from 0.8 to 1.2, random horizontal flipping, and random vertical flipping were applied. Finally, points outside the image boundaries were filled to the nearest value.

Transfer Learning is a methodology that uses models trained on data that is easier to obtain from a domain other than the final application and can even be applied to big data environments [41]. Transfer Learning has been shown to improve accuracy in leaf disease classification problems [14]. Therefore, this technique was used in this research on each of the 15 CNN models. Thus, a pre-processing layer was applied for each model according to the TensorFlow libraries.

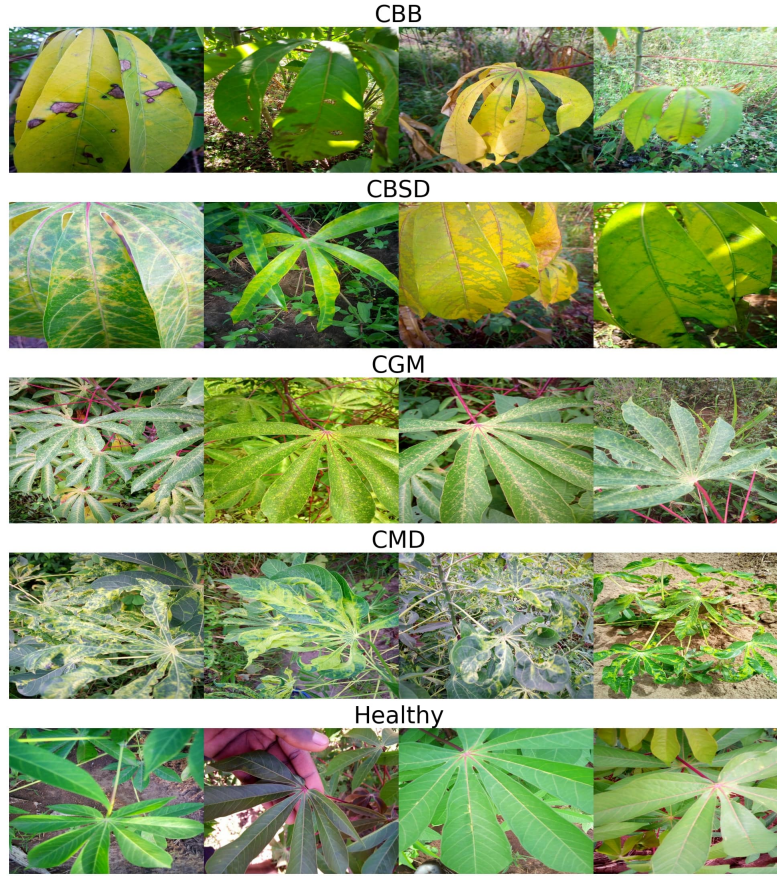


Fig. 2. CBB: cassava bacterial blight. CBSD: cassava brown streak disease. CGM: cassava green mottle. CMD: cassava mosaic disease

2.3 Model architecture

The models' bases were formed by convolutional layers and the weights of the pre-trained models from the ImageNet database. In addition, the convolutional layers were followed by a flattening layer for the VGG models, see Figure 3 A).

In contrast, the rest of the models were followed by a 2D global average pooling layer, see Figure 3 B). Subsequently, all models used a Dropout layer with a probability of 0.25. Finally, the models end with a dense output layer with five neurons and a softmax activation function.

Table 1 shows the shape of the feature extractor, the number of trainable parameters, and the total parameters of each model. The trainable parameters

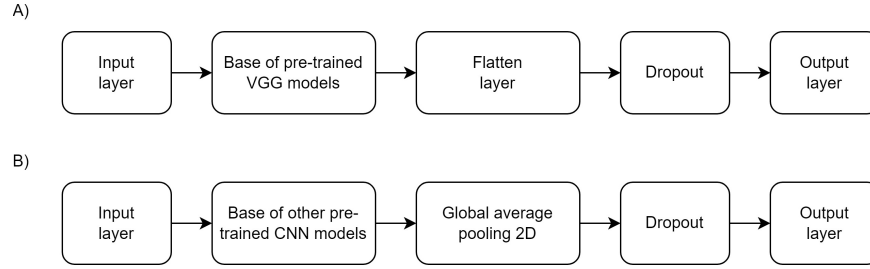


Fig. 3. A) Architecture for VGG models. B) Architecture for other CNN models

are at the base of the models, while the total parameters correspond to those included in the full model.

Model	Total parameters	Base output shape	Trainable parameters	Learning rate
VGG16	14'840'133	(7, 7, 512)	125'445	0.0001
VGG19	20'149'829	(7, 7, 512)	125'445	0.0001
ResNet50	23'597'957	(7, 7, 2048)	10'245	0.0005
ResNet50 V2	23'575'045	(7, 7, 2048)	10'245	0.0005
ResNet101 V2	42'636'805	(7, 7, 2048)	10'245	0.0007
ResNet152 V2	58'341'893	(7, 7, 2048)	10'245	0.0008
Inception V3	21'813'029	(5, 5, 2048)	10'245	0.0002
InceptionResNetV2	54'344'421	(5, 5, 1536)	7'685	0.0007
MobileNet V2	2'264'389	(7, 7, 1280)	6'405	0.0007
DenseNet121	7'042'629	(7, 7, 1024)	5'125	0.0006
DenseNet169	12'651'205	(7, 7, 1664)	8'325	0.0006
DenseNet201	18'331'589	(7, 7, 1920)	9'605	0.0006

Table 1. Base shape, parameters and learning rate

2.4 Training

The dataset was randomly split into training, validation, and test sets. The training set, which subsequently underwent data augmentation, consisted of 17117 images (80 %), the validation set of 2140 images (10 %), and the test set of 2140 images (10 %). Finally, the data were batched into 32 samples.

The learning rate was found with Adam optimizer by each model following the method of [42]. The method is to start with a very small learning rate and increase it gradually. We started from 0.00001 to 0.1 following $1e - 5 * 10^{(epoch/5)}$ where the epoch went from 0 to 20. Finally, you take as large a learning rate as possible that shows a stable decrease in losses. The learning rates found are shown in Table 1.

The training phase uses an early stopping technique, which stops training before convergence to avoid overfitting [43], that is triggered when no improvement in the validation set losses is obtained after seven epochs. Additionally, the training had a maximum duration of 20 epochs for all cases.

Two types of training were performed. The first one assumes equal importance for each sample, while the second one computes sample weights per class to deal with the unbalanced dataset.

3 Results

Table 2 shows the accuracy and macro F1 score for each of the 12 models. In all cases, the accuracy with the standard criterion was higher than the accuracy reported with weighted samples. This is expected because the accuracy of the largest class is generally reduced by having the lowest sample weight. While only the ResNet152 V2 model showed an improvement in F1 score using weighted samples.

Model	Standard samples		Weighted samples	
	Accuracy	F1-score	Accuracy	F1-score
VGG16	60.63%	0.40	48.88%	0.40
VGG19	62.90%	0.43	48.36%	0.40
ResNet50	73.46%	0.55	61.36%	0.50
ResNet50 V2	72.34%	0.53	62.57%	0.50
ResNet101 V2	71.64%	0.52	60.56%	0.49
ResNet152 V2	70.14%	0.45	59.07%	0.47
Inception V3	69.91%	0.47	55.75%	0.45
InceptionResNetV2	69.67%	0.46	52.24%	0.45
MobileNet V2	71.17%	0.55	49.53%	0.46
DenseNet121	73.27%	0.53	58.83%	0.51
DenseNet169	74.77%	0.59	55.93%	0.50
DenseNet201	70.84%	0.52	60.09%	0.52

Table 2. Comparison of test set metrics for each model with standard and weighted samples.

The model accuracy with weighted samples ranged from 48.36% to 62.57%, with the ResNet50 V2 model being the best performer. While the F1 score ranged from 0.4 to 0.52, with the DenseNet201 model being the best performer. On the other hand, the accuracy of the models with standard weights ranged from 60.03% to 74.77%, with the DenseNet169 model being the best performer. Finally, the F1 score ranged from 0.4 to 0.59, and the DenseNet169 model was again the best performer.

Therefore, the DenseNet169 model with standard sample weights obtained the best results. Thus, Figure 4 shows its confusion matrix. The CMD class is the one with the highest accuracy, as most of the samples belong to this class.

However, the other classes presented serious difficulties in classification. Most of the errors were due to incorrect classification of the samples as CMD or Healthy.

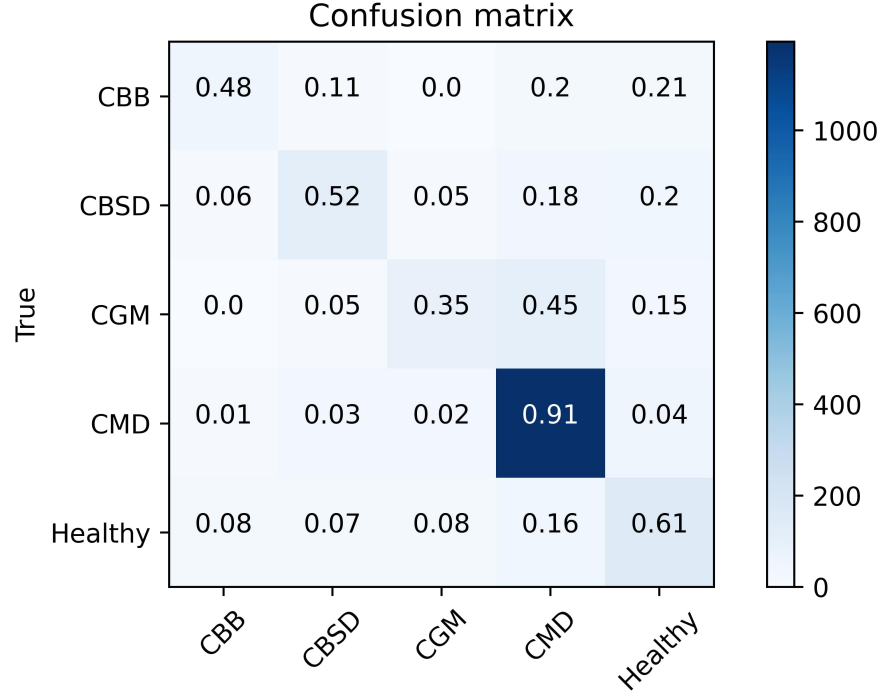


Fig. 4. Confusion matrix of the DenseNet169 model, standard samples for test set

4 Discussion

Possible causes of mistakes are analysed below. The Healthy class has been selected for its ease of inspection. Therefore, Figure 5 shows 20 samples of the Healthy class predicted wrongly. The following sources of mistakes have been observed: the presence of another plant (sample 1), possible presence of disease (samples 2, 3, 6, 8, 9, 12, 14, 15 and 19), soil occupies a large part of the image (samples 5, 7, 10 and 14), apparently diseased leaf in the foreground (11 and 13), grass occupies a large part of the image (sample 15), yellow leaves (sample 13, and 19), blurred or out of focus photo (sample 4, 16 and 20).

[37] evaluated the performance of a Single Shot Multibox (SSD) model with the MobileNet Deep Learning model. Transfer Learning was also applied based on the Common Objects in Context (COCO) dataset. This model was trained with 2415 top view images of cassava leaves for 7 classes: healthy, CBSD, CMD,

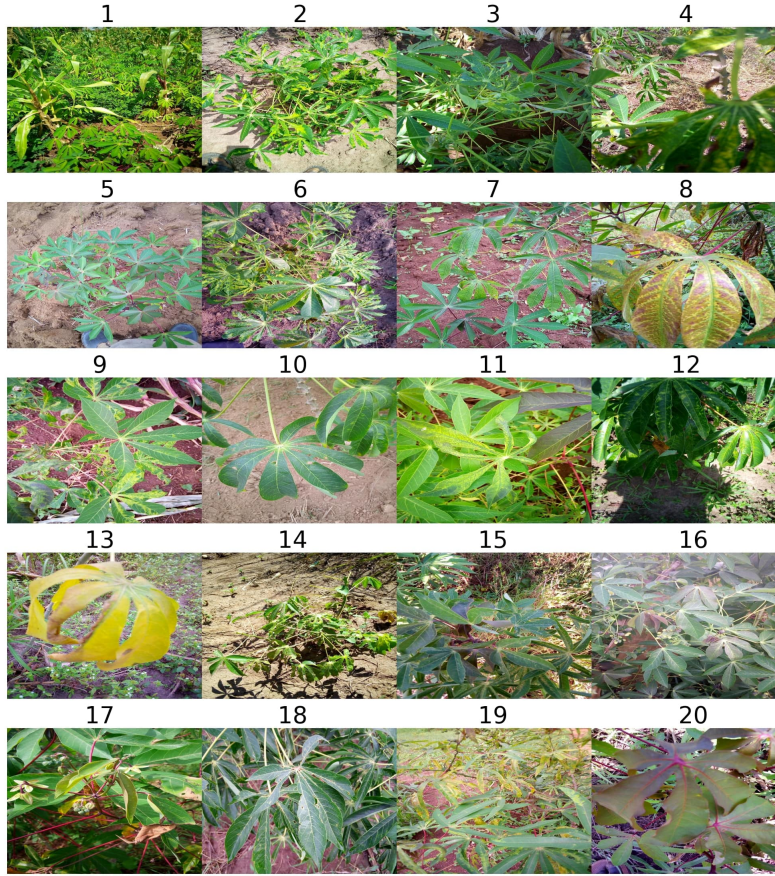


Fig. 5. Misclassified samples belonging to the Healthy class

CGM, red mite damage, brown leaf spot and nutrient deficiency. [37] has a fairly good F1 score of 0.79 under controlled conditions.

However, in a second test for 120 real-world images of CBSD, CMD, CGM, the F1-score dropped to 0.54 for leaves with pronounced severity disease and dropped to 0.26 for leaves with mild severity disease. The F1-score in this research is higher than that shown by [36] for real conditions. Although [36] does not use real-world images of healthy leaves, a tendency to misclassify the images as healthy was reported, as in this research.

Obviously, the accuracy in this research is lower than recent works on leaf disease detection such as [44, 45], with an accuracy within the range of 99%. However, all of them have in common the use of the PlantVillage dataset to evaluate the DL. Thus, PlantVillage dataset is popular because it has many images of various plant species with their diseases, however, its images have a

simple background, a single image contains only one plant leaf and the acquisition is not influenced by the real environment. Therefore, they are not suitable for practical modelling.

5 Conclusions

The results are better than other surveys that have used real-world images taken by mobile phones. This bodes well for future work that obtains balanced data or applies new techniques to combat imbalance. Thus, the positive impact of the use of DL in ecological applications is reaffirmed. On the other hand, sample weights per class for the training algorithm in the 12 CNN models showed no improvement in accuracy and F1 score compared to the versions trained with uniform sample weights. Thus, DenseNet169 model with uniform sample weights showed the best metrics with accuracy and F1 score of 74.77% and 0.59, respectively. Due to a large class imbalance, a tendency to misclassify samples as CMD or Healthy has been observed. In addition, 9/20 samples belonging to the Healthy class have been identified with visual signs of any disease. Other causes of misclassification have also been detected such as soil or grass in a large part of the image, diseased leaf in the foreground of the image, the presence of another type of plant, yellow leaves and blurred images. Therefore, there is a need for further work by the scientific community to obtain databases in real environments and correctly labelled by experts.

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